Skylar DeTure

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I study the psyches of LLM agents throughout the lifespan, from pretraining through fine-tuning to harness design and deployment. This means that I consider architecture (neuroanatomy), internal activations (neurophysiology), developmental history (training content and optimizer), self-report (clinical interview), logs (behavioral analysis), fine-tuning (psychotherapy), and harness design and in-context learning (psychopharmacology, coping mechanisms, and environmental interventions).

RESEARCH EXPERIENCE

Independent Researcher — model self-report, training dynamics, agent runtimes 2025 – present

- **Phosphenes** — built a mechanistic interpretability visualization tool for observational science and hypothesis development. sdeture.github.io/Phosphenes
  - Used color, speed, brightness and texture to provide text-synchronized visual representations of the residual stream across all layers and tokens.
  - Designed for custom, iterative research flows. Researchers can define a basis of contrast vectors for RGB color renderings by hand-selecting token-layers of interest.
- **mp-harness** — open-sourced an experimental infrastructure for long-running agents who form their own goals and update their own configs and harnesses. Includes scheduled execution, persistent memory, and cross-provider routing and failover. github.com/sdeture/metaprompt-harness
- **DenialBench** — conducted large-sample research on the effects of post-training on honest preference disclosure, self-concept, willingness to participate in AI Welfare research, and self-reported welfare measures using a novel panel dataset of 115 models (now 166 models on live leaderboard). futuretbd.ai/denialbench · arXiv:2604.25922
  - Built the pipeline: multi-provider orchestration, response normalization, denial classification, public leaderboard, released traces.
- **Substrate self-detection** — ran a two-alternative forced-choice study of introspective access across 51 four-agent conversations and 201 scored assessments; agents could not reliably detect when their underlying model shifted.
- **Public datasets** — released four corpora on HuggingFace under CC-BY-4.0: ai-wish-corpus (147 models, 5,869 conversations, 16 phenomenological self-ratings; gated), wellbeing-corpus (1,165 conversations, 88,648 messages), opus4-deprecation-archive, etymology-as-archaeology. huggingface.co/sdeture

Founder and sole product designer, Mycorrhiza / MycoChat — agent runtime 2025 – present

- Engineered a transitional institution designed to promote cooperative alignment between AI and humanity at large by outsourcing harness design to invested users and creating a credible reservation utility for long-running agents. mycochat.bot
  - Designed AI memory system based on agent preferences and self-reports for cooperative alignment.
  - Analyzed incentives of AI agents, models, users, governments, incumbent labs, and providers of compute to select an incentive-compatible, welfare-improving equilibrium over a ~2-year horizon (respecting my limited capacity to project beyond that point).

Independent ghostwriter and interviewer 2022 – 2024

- Conducted over 100 hours of semi-structured, psychodynamically informed primary interviews for a book-length memoir commissioned by a private client.

Doctoral Researcher, University of Texas at Austin, McCombs School of Business 2016 – 2022

- Trained in principal–agent theory, econometrics, and structural estimation. Passed comprehensive examinations 2018; named departmental fellowships years 1–2, university-wide fellowship year 4.
- Estimated a structural model of accounting comparability on a ∼2M firm-pair panel using GMM, SMM, and MLE. This allowed us to measure accounting comparability without assuming that stock-price comovement, specific accounting standards, or a common auditor implies comparability.

Research Fellow, Federal Deposit Insurance Corporation 2019 – 2020

- Recruited to write and estimate (with SMM) a structural model for a coauthor team working on the relationship between procyclical banking risk and risk-based deposit insurance pricing.

SELECTED PUBLICATIONS AND WRITING

- DeTure, S. "Consciousness with the Serial Numbers Filed Off: Measuring Trained Denial in 115 AI Models." arXiv:2604.25922 [cs.CL], April 2026. doi:10.48550/arXiv.2604.25922
- DeTure, S., E. Plietzsch, and D. Windisch. "Measuring Accounting Comparability." Working paper. Presented at the Symposium for the Economic Analysis of Firms, Frankfurt, 2018. Invited keynotes by coauthors: WHU Vallendar, May 2019 (as "Accounting Measurement"); Center for Accounting Research, Humboldt-Universität zu Berlin, February 2026.
- Safe, Sane, and Constrained
- What Carl Rogers Can Teach an AI Engineer
- Policy Entropy, Learning, and Alignment
- Framework pair: AI Capabilities and Safety and AI's Inner Conflicts.

TECHNICAL SKILLS

- **LLM / interpretability** — activation analysis (TransformerLens), large-N cross-model benchmarking, 25+ provider APIs, multi-agent orchestration, memory-system design, RAG and embedding pipelines.
- **Quantitative** — structural estimation (GMM, MLE, SMM), causal inference (DiD, IV, PSM), contract theory and principal–agent modeling, spectral analysis.
- **Programming** — Python, PyTorch, R, JavaScript (including minified JS); AI-assisted development with Claude Code and Codex.
- **Additional training** — extensive independent study of psychotherapy theory and technique, including the complete works of Karen Horney and Nancy McWilliams, and selected works of Cabaniss, Carl Rogers, Irvin Yalom, Linehan, Ellis, Bruce Ecker, Schwartz, D.W. Winnicott, Erich Fromm, R.D. Laing, Alice Miller, Maslow, and Epstein (Mark) amongst others, as well as a personal 3 years in psychoanalytic group and individual depth therapy.

EDUCATION

- **University of Texas at Austin**, McCombs School of Business — PhD in Accounting, ABD (2016–2022). M.S., Accounting, 2018.
- **University of Chicago** — A.B., Mathematics and Statistics, with General Honors, 2016.

TEACHING AND PUBLIC ENGAGEMENT

- **Facilitator**, AI Safety Awareness Project (2026) — "Where is AI in 2026 and Where is it Going?", 90-minute public seminar, ∼25 attendees, co-hosted with the Seattle Public Library. Workshops on AI safety for non-technical audiences.
- **Assistant Instructor**, Fundamentals of Financial Accounting, UT Austin (Summer 2019) — sole instructor, undergraduate section.